

Microservices Using Node/ExpressJS with ES6+ on Kubernetes



NodeJS is one of the most popular platforms for developing microservices. The ECMA standard for modular, object-oriented and functional programming made JavaScript a promising language. Many enterprises choose the ExpressJS framework for exposing the microservices as REST API on the *Node.js* platform. In this Part 11 of the series, we explore this stack by developing the *SearchService* of the UMS and deploying it on Kubernetes in such a way that the *SearchService* co-lives and collaborates with the other services, with scale.

We have already developed the *AddService* on the Java platform using the Spring framework and *FindService* on the Python platform using the Flask framework. We did it this way just to convince ourselves that it is absolutely possible to design, develop and deploy collaborating microservices on a variety of platforms. This time, we choose NodeJS as the third and last platform to explore microservices development.

SearchService

This microservice offers an API for searching users in the UMS database by phone number. Since the UMS does not insist on unique phone numbers, the API accepts phone as a query string parameter.

```
GET /user?phone={PHONE}
```

The API returns a JSON array of zero, one or more users against the given phone number.

The design for the service was already presented in Part 5 of this series. Let us implement the design using

the ExpressJS framework and MySQL connector on the NodeJS platform.

Project setup

Like with any other NodeJS application, the process starts with initialising the project. Once the Node and NPM are installed, the following interactive command creates the basic project structure in the current folder:

```
npm init
```

A successful run of the above command generates a file named *package.json*. Edit the file to add the scripts and dependencies among other things.

```
{
  "name": "ums-search-service",
  "version": "1.0.0",
  "description": "search service for ums",
  "main": "./src/index.js",
  "type": "module",
```